8

Chapter 8 - Case Study: A sourceForge project, Technology Explorer for IBM DB2

Some open source projects grow along predictable lines. Some start well but quickly lose momentum. Just look at the number of projects on sourceforge.net that have had no recent activity. Other projects grow organically in unpredictable directions. The Technology Explorer for IBM DB2 or TE for short (http://db2mc.sf.net) is a project that has been through a long evolution. The TE started like all good open source projects, as an itch that needed scratching.

8.1 What is the Technology Explorer for IBM DB2?

The Technology Explorer is a light weight, Web based console for DB2 for Linux, UNIX and Windows. The Technology Explorer is also a teaching and exploration tool for DB2. It is a great way to learn about DB2 because it doesn't just explain how to do things; it lets you play using a live database. The TE includes hundreds of ways to see inside the database and monitor what is going on. It also includes numerous tutorials and advanced techniques to get the most out of your system.

The Technology Explorer is meant to be a tool for the community. You can use it, change it, extend it and use it to share your ideas. Everything is defined using XML so you don't need to write code to make changes.

Some of the key features of the Technology Explorer are:

- Is a light weight Web based platform for interacting with DB2 for Linux, UNIX and Windows
- Is easy to expand and customize
- Works with DB2 for Linux, UNIX and Windows version 9.1, 9.5 and 9.7
- Connects to any DB2 data server using only an IP address
- Contains a wealth of content to highlight, demonstrate and teach you about DB2's core features

Figure 8.1 shows the Technology Explorer. It can run on most popular browsers.

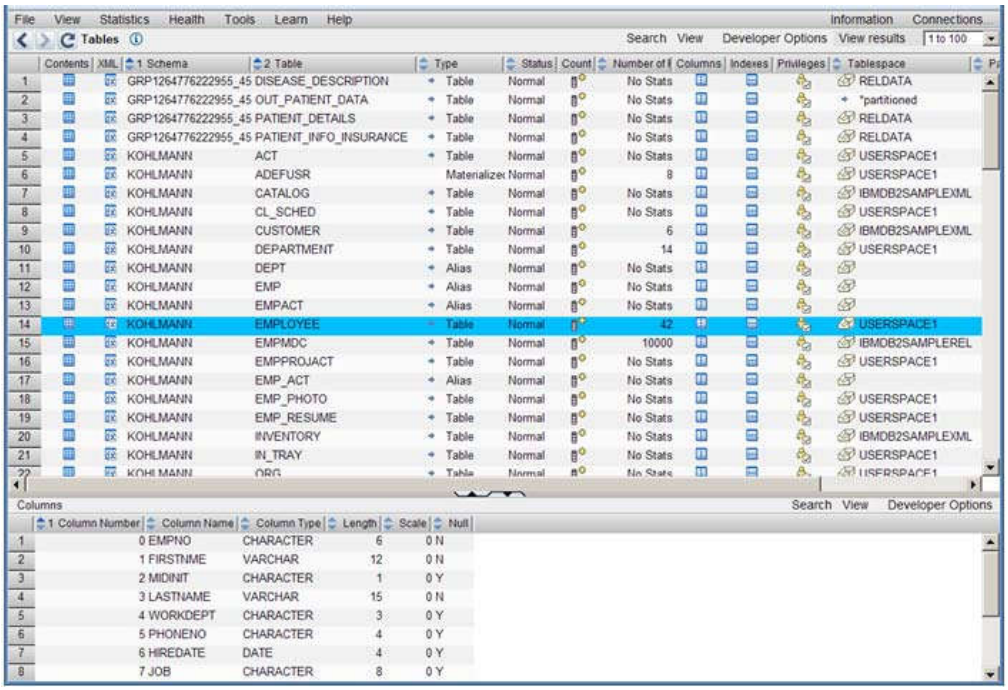


Figure 8.1 – The Technology Explorer for IBM DB2

8.2 A quick overview of the Technology Explorer for IBM DB2

This section provides a quick overview of the TE. You will learn the requirements to set it up, and its basic operations. The easiest way to get started wit the TE is by looking up its homepage on SourceForge http://sourceforge.net/projects/db2mc/

This site includes everything about the TE including setup instructions, wikis, requirements and forums as well as the source code.

8.2.1 Requirements for setting up the TE

To start working with the TE, you will need to:

- Install DB2
  - Download DB2 Express-C, the free version of DB2 from **http://www.ibm.com/db2/express/download.html**
- Install Apache
  - Download Apache Web server from **http://httpd.apache.org/**
- Install PHP
  - Download PHP from **http://www.php.net/**

- Install the IBM DB2 driver for PHP
  - Download this driver from **http://sourceforge.net/projects/db2mc/files/**
- Download and unzip TE
  - Download the TE from **http://sourceforge.net/projects/db2mc/**

For more information about DB2 refer to the free eBook *Getting Started with DB2 Express-C* that is part of the DB2 on Campus book series. Alternatively, you can also review *Appendix B, Up and Running with DB2*.

For instructions on how to set up Apache and PHP for TE refer also to the TE page on SourceForge. The development team is working on simplifying the set-up process. At the time you are reading this, a single package installer may already be available on the TE SourceForge site.

8.2.2 Some basic features and operations of the TE

This document will briefly discuss some basic features and operations that the TE offers:

- Connect to a database
- Display table and view contents, and the output of queries
- Run ad hoc queries
- View and change database configuration and monitoring parameters
- Display basic monitoring information
- Create and work with tutorials and workshops

If you would like to know more about how to use TE in detail, a step-by-step guide is also available on the TE SourceForge site.

8.2.2.1 Connecting to a Database

Figure 8.2 shows the TE starting page. This is the page that appears when you start the TE in your browser.

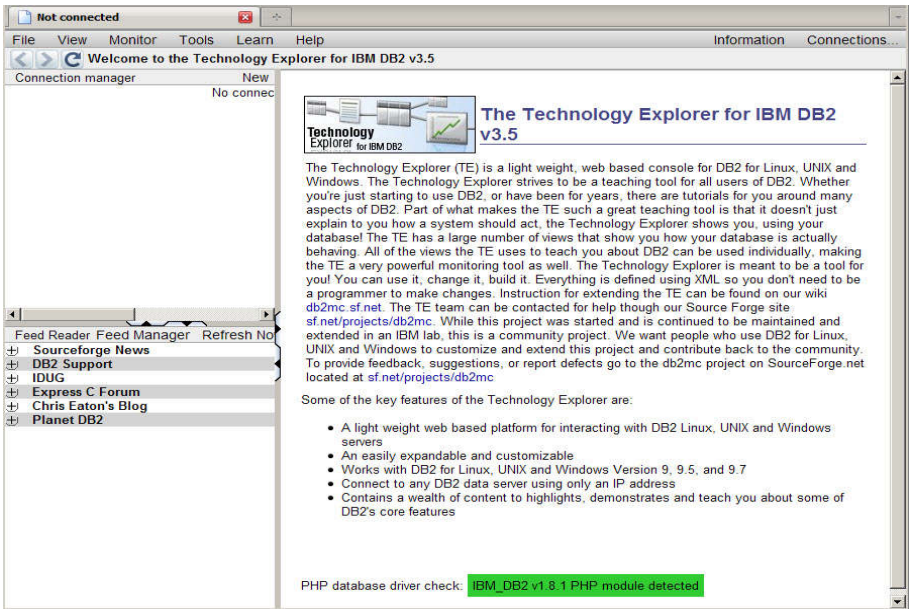


Figure 8.2 – TE starting page

The main menu bar is at the top of the page. The database connection manager is at the top left. The RSS feed reader is at the bottom left; and the introduction page is on the right.

To create a new database connection, click on *New* in the top right corner of the connection manager as shown in *Figure 8.3* below.



Figure 8.3 – Creating a new connection

If you are connecting to a database **on your local machine**, enter the database name, user name, and password, as shown in *Figure 8.4* below. The password field color will change from red to green when you enter a password.

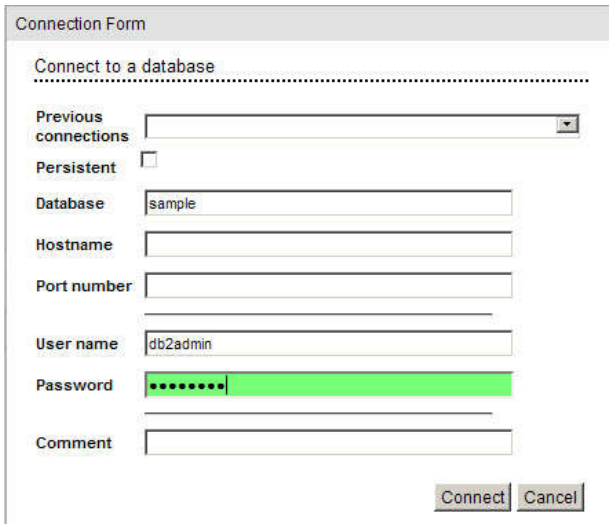


Figure 8.4 – Connection Form

If you are connecting to **a remote database**, then you must also enter the host name of the database machine as well as the port number used by the DB2 instance (the default is 50000). Click on *Connect*. The TE should load with the connection details in the top left panel, with an arrow under the *Active* field, and a check mark under the *Authenticated* field as shown in *Figure 8.5* below.

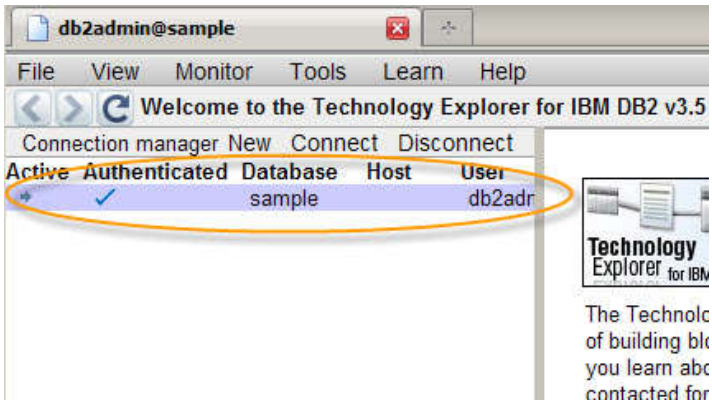


Figure 8.5 – Connected to the Sample database

If you get an error message, check that:

- You have entered the correct combination of user name and password
- The user has authorization to connect to the database
- The database name is spelled properly
- For remote databases, make sure the host name and port number are correct
- The DB2 database server is running

Note:

Connecting to a DB2 database is discussed in detail in the free eBook *Getting started with DB2 Express-C*. You can also watch this video:

http://www.channeldb2.com/video/video/show?id=807741%3AVideo%3A4222

8.2.2.2 Display tables, views, and aliases

With the TE, you can display DB2 catalog tables, and views. Select *DB2 catalog views* from the *View* menu at the top of the page as illustrated in *Figure 8.6.*

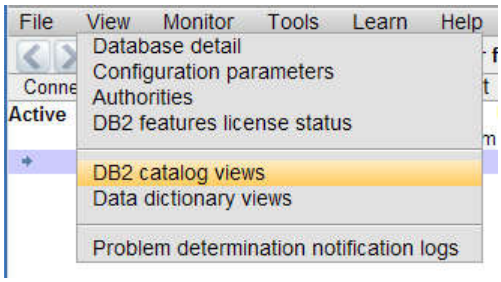


Figure 8.6 – Opening the Catalog Views

Figure 8.7 below provides an example of all items in the SYSCAT.TABLES view.

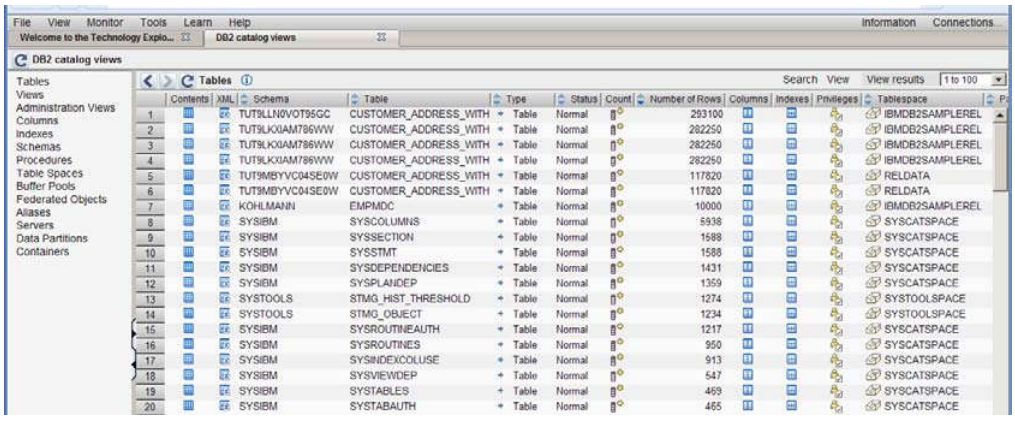


Figure 8.7 – SYSCAT.TABLES catalog view

From this page you can see the query that retrieved the data in the table by clicking on *View -> Profile Query* as shown in *Figure 8.8*.

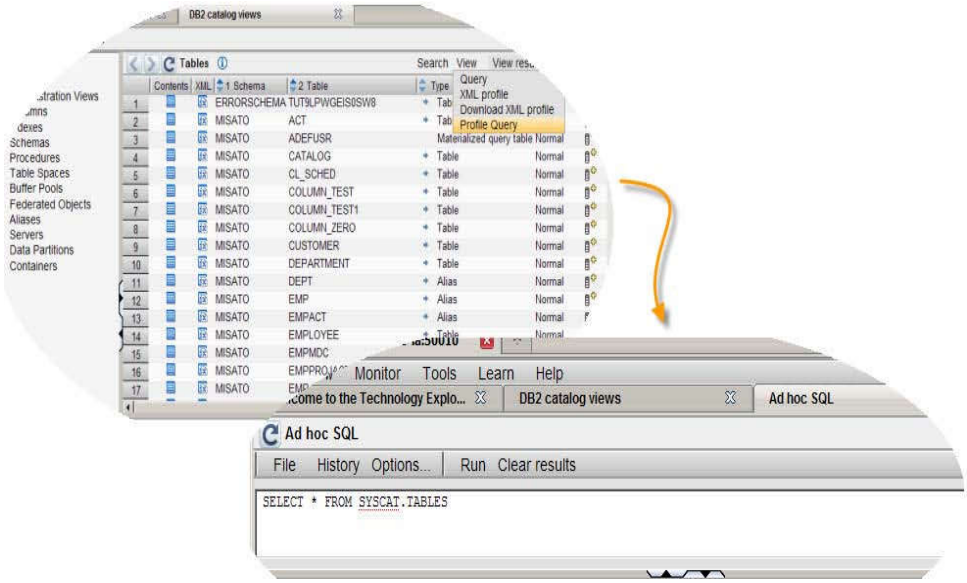


Figure 8.8 – Viewing the query that retrieved the data displayed on the page

You can also see the contents of each table by selecting the *See the contents* icon at the left of the display. A new tab opens with the contents of the table or view youselected. This is illustrated in *Figure 8.9* below for table DB2INST2.ACT.

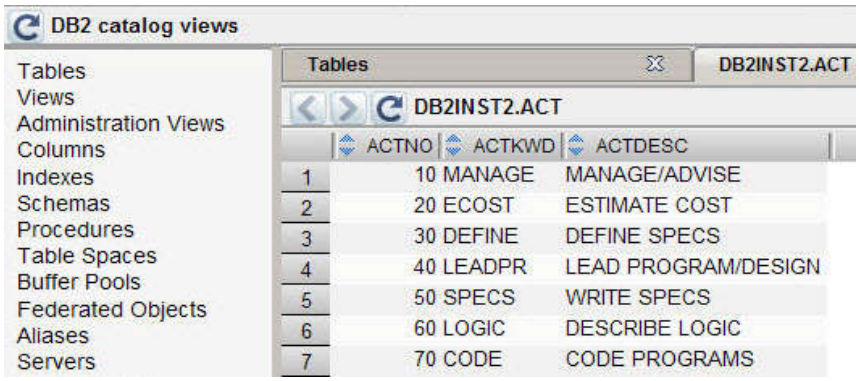


Figure 8.9 – Contents of a table

8.2.2.3 Running ad hoc queries in TE

TE allows you to run SQL statements. You can see the results including BLOB, CLOB, DBCLOB, and XML data. *Figure 8.10* provides an example of an SQL statement (at the top) and the output (at the bottom). In the example the RESUME column is defined as

CLOB. On the left side of the result, there is an icon that when clicked will display the CLOB contents on the right for the given employee.

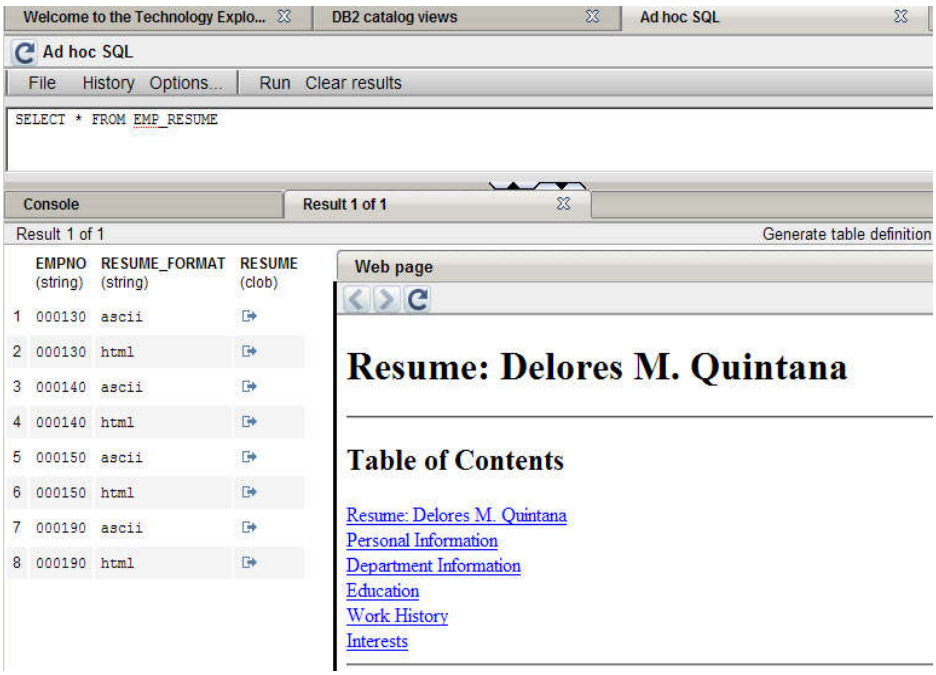


Figure 8.10 – Running an ad hoc query

8.2.2.4 View and change database configuration and monitoring parameters

The TE allows you to change database configuration and monitoring parameters without having to revert to the command line. The icons highlighted in *Figure 8.11* below are used to change values.

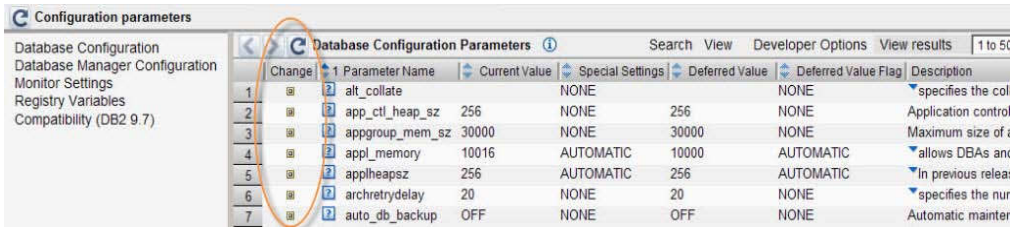


Figure 8.11 – Changing database configuration parameters

8.2.2.5 Display monitoring information

The TE includes monitoring information under the *Monitor* menu. DB2 can provide detailed monitoring information through SQL, which allows TE to show you what is happening in your DB2 server. *Figure 8.12* provides a sample of a monitoring dashboard. You can set it to refresh as often as you like.

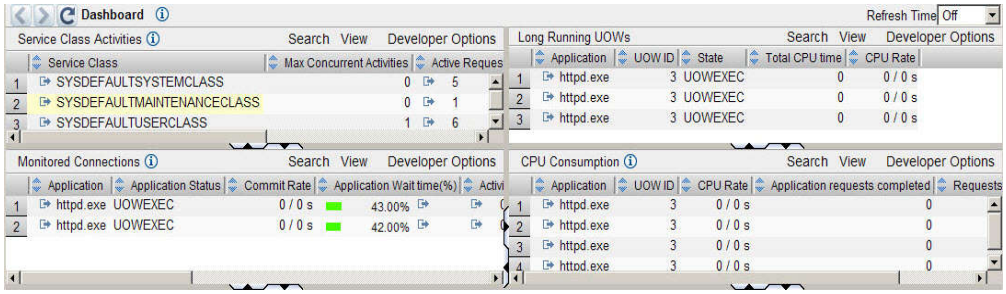


Figure 8.12 – The monitoring dashboard

8.2.2.6 Tutorials/Workshops

The TE offers many tutorials and workshops on DB2 as well as on the TE itself. *Figure 8.13* shows a small sample of the DB2 feature-based tutorials available. If you want to learn how to customize the TE, have a look at what is available under *Learn -> Workshops on Technology Explorer* menu.

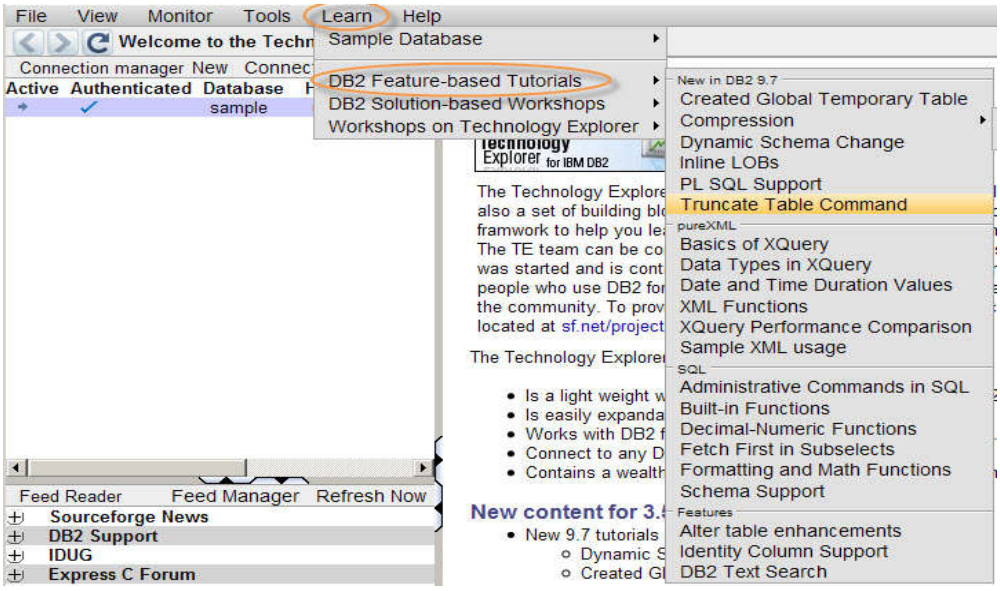


Figure 8.13 - Tutorials and workshops available with TE

8.3 You need a key insight to build a project

The TE started as a hobby project and a way to explore DB2 and PHP. It started with basic information from a developerWorks® article (http://www.ibm.com/developerworks/data) on how to connect PHP to a DB2 data server. By writing a set of functions to encapsulate connection information and process SQL it was easier to quickly retrieve data from the server. Then by creating a set of PHP objects that describe tables and how to display them it became easy to quickly display data in a standard way. The first version was a sample PHP application for the sample database shipped with DB2. It demonstrated some basic principals of good PHP application development and examples of how to interface with DB2.

It evolved into an Open Source project when we understood the importance of flexibility in the design and the value of engaging the DB2 community. The first important insight was that sample data provided with the DB2 Sample database isn't as interesting as system data. The sample data serves no immediate practical purpose. We needed something with a practical application that would attract users to the project.

Fortunately DB2 9.1 has a ready source of data that is useful to anyone using a DB2 database - monitoring and performance data. By switching from a sample business application to a sample monitoring and management application the ***DB2 Monitoring Console Open Source Project*** was created (if you were wondering why the URL for the project is db2mc).

The second insight was that you could describe database tables and views using XML files. By switching to XML we created a far more flexible development environment. It became easy for anyone to customize the db2mc without writing a single line of code. We also took advantage of the outstanding XML support in IBM DB2. By using an XQuery statement we could automatically create a default *XML profile* for any table or view directly from the DB2 catalog with just a push of a button. Creating a new display used to take an hour or more. It now took minutes. *Figure 8.14* shows an example of an XML profile.

```
<table name="KOHLMANN.DEPARTMENT">
        <sql_name>&quot;KOHLMANN&quot;.&quot;DEPARTMENT&quot;</sql_name>
        <singular_name>KOHLMANN.DEPARTMENT</singular_name>
        <plural_name>KOHLMANN.DEPARTMENT</plural_name>
        <description/>
        <rows_per_page>50</rows_per_page>
        <order_by_index/>
        <display_columns>
                <col type="column" name="deptno"/>
                <col type="column" name="deptname"/>
                <col type="column" name="mgrno"/>
                <col type="column" name="admrdept"/>
                <col type="column" name="location"/>
        </display_columns>
        <column name="deptno">
                <c_sql_name>&quot;DEPTNO&quot;</c_sql_name>
                <c_title>DEPTNO</c_title>
                <c_presentation>s</c_presentation>
                <c_drill>y</c_drill>
        </column>
        <column name="deptname">
                <c_sql_name>&quot;DEPTNAME&quot;</c_sql_name>
                <c_title>DEPTNAME</c_title>
                <c_presentation>s</c_presentation>
                <c_drill>y</c_drill>
        </column>
        <column name="mgrno">
                <c_sql_name>&quot;MGRNO&quot;</c_sql_name>
                <c_title>MGRNO</c_title>
                <c_presentation>s</c_presentation>
                <c_drill>y</c_drill>
        </column>
        <column name="admrdept">
                <c_sql_name>&quot;ADMRDEPT&quot;</c_sql_name>
                <c_title>ADMRDEPT</c_title>
                <c_presentation>s</c_presentation>
                <c_drill>y</c_drill>
        </column>
        <column name="location">
                <c_sql_name>&quot;LOCATION&quot;</c_sql_name>
                <c_title>LOCATION</c_title>
                <c_presentation>s</c_presentation>
                <c_drill>y</c_drill>
        </column>
</table>
```

Figure 8.14 – XML Profile

This also added the ability to look at the contents of any table or view. Selecting the *contents* icons on the tables or view catalog page generates a default profile that is fed back to the page generator to create a default display for any page or view. This eliminated the tedious task of creating new pages.

With this new flexibility we also wanted to start engaging the DB2 community. Quick and simple displays are only useful if they contain meaningful information. The best way we felt to do this was to make the code available as an open source project. We hoped that DBA's would take some of the SQL queries they use every day and convert them to pages in the db2mc. More importantly we wanted them to contribute their work back to the DB2

community so other people could see the insight they gained from SQL that some DBA had taken years to develop.

The key insight that drives the Technology Explorer project is that to engage the community of database users you need an interface that is as flexible and dynamic as the SQL or XML languages. There are over 200 XML profiles shipped with the TE. They not only account for over 90% of the code shipped with the TE they represent an open language that people in the community can use to communicate their ideas of how they work.

8.4 You need to support and grow a community

Early adopters and contributors are the lifeblood of any open source project. You ignore them at your peril. For every 100 or 1000 users you are lucky to find one contributor. We were lucky enough that a very experienced and knowledgeable DBA found our project. Peter Prib (http://sourceforge.net/users/peterprib) has become an exceptional contributor to the project. He has added numerous new pages, new tools and utilities, and even more branches of code that we have yet been able to integrate into the mainline code stream. If he needs something to support his database administration consulting practice he adds it to the project. He has also been a great tester and regularly opens defects and adds requirements. Peter Prib's most important role is as our domain expert. He works with DB2 servers and customers every day. There is no substitute for that kind of practical experience. He doesn't just say that something is important, he invests his time.

We also receive contributions from universities and colleges. They are an outstanding resource for contribution. Working on an open source project not only gives a student practical experience but can become part of their portfolio of work. Most students work on software owned by another company. They can't show their code to anyone else or even prove that their code was used in the final product.

Contributing to an open source project has two advantages. First someone else usually approves your code and may give you valuable feedback on your style and technique. And second your code is available for anyone to see. A prospective employer can see for themselves that you have experience building code that was accepted by your peers.

The Technology Explorer for IBM DB2 uses a very open Apache license that makes it easy for anyone in the community to contribute. If you choose to contribute to a project, don't be surprised if they ask you to sign a contributor's agreement. They just need proof that the code you are contributing is original.

8.5 Make your project easy to adopt

Adoption isn't usability. It is much more than that. People go through a very well defined set of steps when they adopt new technology. As Jonathan Schwartz, from Sun Microsystems, has said, "*Adoption is a non-economic phenomena, no money is spent, only time…*" In other words, people spend time when they investigate new technology before they spend any money. Even if open source is free, if you want people to adopt your project, you need

to make their initial time investment very low. You need to make your project convenient to use.

The first step is to ensure the software is easy to install and try out. If possible, build a live Web site where people can try out your project without installing anything. We made sure that we could set up the Technology Explorer for IBM DB2 so it could run in demonstration mode. Take a look at http://db2mc.dfw.ibm.com for an example. The setup ensures that anything destructive has been turned off.

Invest the time in an integrated installer. The only clearly negative feedback on the TE is that it takes too many steps to set up. A common problem with open source projects is that they are built on several components. The TE requires a Web server, DB2 PHP drivers as well as the TE code. For someone who has been through the process before, or is familiar with Web applications, it only takes five minutes, but to a novice, the number of steps may seem complex. We are finalizing an integrated installer that lays down all the components along with a single desktop icon to launch the TE and hope to have legal approval to ship all the components by the time you read this.

The second step is to ensure that someone can quickly see how to adapt the software to their own needs. In the more recent versions of the Technology Explorer we included demonstrations on how to customize the system including detailed examples. *Figure 8.15* illustrates a tutorial about how to customize the TE.

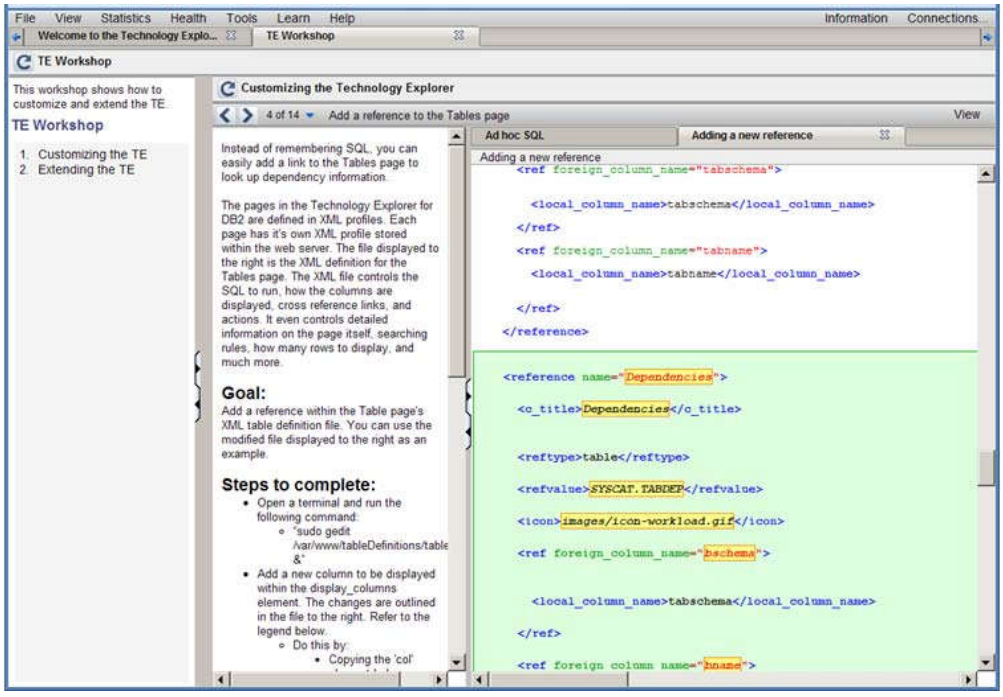


Figure 8.15 – Tutorial on how to customize the Technology Explorer

Adopting is also about easily making the transition from what you are already using. As mentioned before, you can easily create a new display in the Technology Explorer by selecting the XML icon from the table or view page. You can also export the SQL embedded in any XML profile to the ad hoc SQL page. Often we found that the results we needed were not available from a simple table or view. DB2 DBAs have also spent years developing SQL to answer key questions about the databases they are managing.

To give them a way to use their SQL we added the ad hoc SQL page. It not only allows them to reuse their SQL; they can quickly build new pages based on their existing SQL. The *Generate table definition XML* actions on the results tab will automatically generate an XML profile for any statement you can run in the ad hoc window. Building a bridge from existing practice to your new technology will greatly accelerate adoption.

Figure 8.16 illustrates running ad hoc statements in the TE.

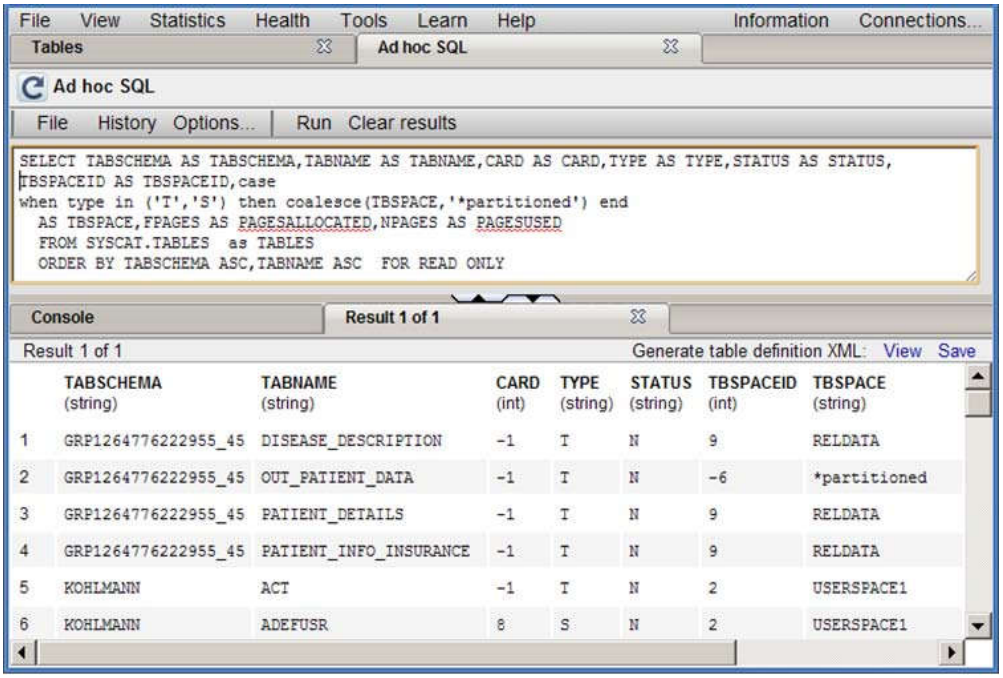


Figure 8.16 – Running ad hoc statements

8.6 Understand your business model

A business model isn't always about making money by selling a product or service. Sometimes, especially with open source, it can be about saving money or sharing a cost and risk. It is always about understanding why you are investing in a project. In some cases open source is the basis for a services business. The software is free, but support comes at a cost.

Several companies are built around this model for Linux or PHP. In some cases there is a problem that is simply more cost effective for many people to share; Apache and Eclipse are good examples. In other cases it is about establishing a programming model or platform to deliver services, Google's Chrome for example.

In the case of the Technology Explorer for IBM DB2 we saw an opportunity to build a business model on demonstrations and tutorials. IBM was looking for ways to deliver more effective live tutorials and demonstrations for the DB2 data server while expanding their open user community and expanding their presence in universities and colleges. The flexible platform based on XML was the perfect starting point.

One of the first things we did to turn the TE from a sample program into a teaching tool was to integrate the DB2 Information Center (http://publib.boulder.ibm.com/infocenter/db2luw/v9r7) into the program. Every page in the TE includes an information icon that explains where the information for the page comes from. *Figure 8.17* below provides an example where you can see that the table page is generated from the SYSCAT.TABLES catalog view. In the *Database Configuration Parameters* page we included information links to the Information Center for each parameter.

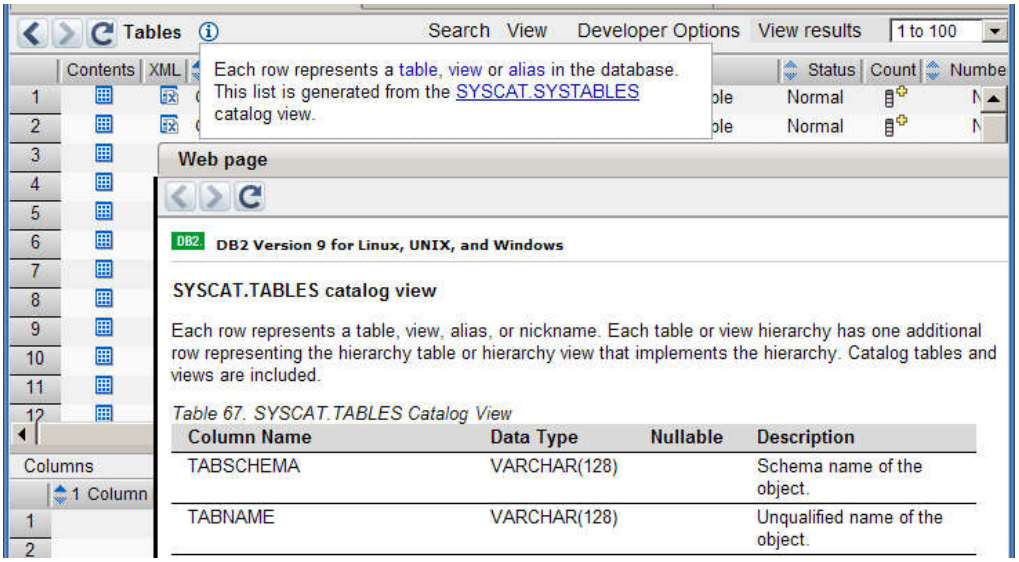


Figure 8.17 –Integrated access to the DB2 Information Center

At the time we also had access to software and paper examples and tutorials. Many based on SQL examples. The biggest problem was that it was inconvenient for people to copy and paste the examples from an online source or worse to manually type from a paper during a workshop.

We extended the XML model to include step by step tutorials. Each tutorial included multiple steps each with a description to the left with a live ad hoc window and results screen to the right. Important parts of the SQL or XML can be highlighted in the ad hoc window. This is illustrated in *Figure 8.19*.

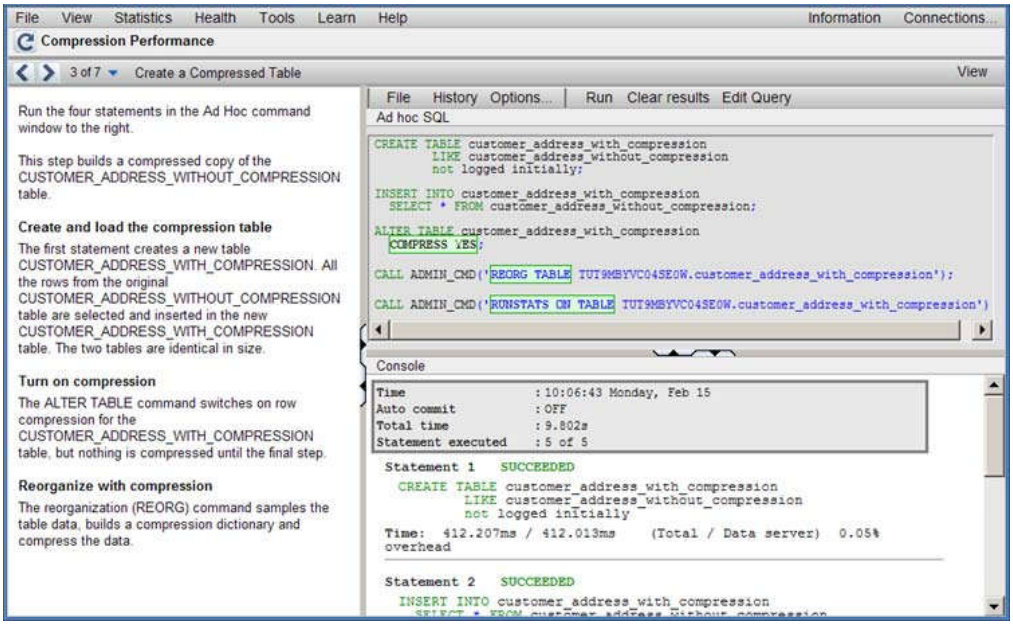


Figure 8.19 –Highlighting SQL

Once you have run through the tutorial, you are free to go back and alter the SQL or XML, and rerun the statements, all against a live database. We also support including pages from the DB2 Information Center or other pages, for example a list of tables, in the right side of the page instead of the ad hoc page.

As the number of tutorials grew, they were grouped together into larger workshops that focus on business problems. Additional functions like graphing and driving multiuser workloads were added to demonstrate more complex scenarios.

The XML language, which we developed to describe each tutorial, acts as a script. It is easy to modify and easy to copy to create a new tutorial. More importantly, just like the XML that defines the display pages in the TE, the tutorial XML acts as a language that people can use to describe their ideas. Instead of simply publishing documents, the tutorial structure can be used to explain a management process to a new DBA or highlight a valuable technique for getting the most out of DB2.

Make sure you take the time to understand why you are investing in an open source project. It may be as simple as a way to learn a new technology. With each step, the reason to keep investing time and resources may change. Whether it was as a sample PHP program, an open source monitor, or a community tool and teaching platform the TE project had a clear set of goals, stakeholders and investment. Throughout we have stayed focused on the original insight that started the project.

8.7 Keep your project current

Do everything in the open. Remember that source forge ranks projects based on community activity as much as it does by downloads. Even if you are working with a small group of contributors open a defect for every change. Track each improvement through a requirement and post your ideas to the forums. The more transparent your work the more active the project will appear and the better your ranking.

We have never advertised the TE. Most new users I talk to say they hear about it through word of mouth or through a lucky search on Google. Blog postings, webpage links, and other references bring most people to the Web site. Of the people who come to the TE sourceForge site about half stay and read about the project for at least an hour.

If you can find an insight to build a project around, make sure it is convenient to use and engages a larger community. Don't lose focus on why you are investing your time at each stage of the project and always do your work in the open.